

Basic Malware Triage Report

PureLogs Infection from Email Attachment

Cybersecurity Portfolio Project

Case Source	Malware-Traffic-Analysis.net - 2025-05-12 PureLogs infection from email attachment
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Report Type	Basic Malware Triage / SOC Portfolio Report
Primary Evidence	Email artifact, PCAP traffic, IOC list, malware/artifact package
Severity	High

Title:

Basic Malware Triage Investigation - PureLogs Infection from Email Attachment

Scenario:

A suspicious invoice-themed email attachment was delivered to a user and led to a malware infection chain. The observed chain was: email -> attachment -> extracted executable malware file. The email subject was "RE: Urgent: Confirmation Required for Invoice & Down Payment Details", and the suspicious attachment was named invoice_10988.xz. The extracted executable associated with the infection was KTMBE25040170.exe. The investigation focused on validating the file, reviewing network traffic, extracting indicators of compromise, and assessing endpoint risk.

Tools used:

- Malware-Traffic-Analysis.net case files
- Wireshark
- VirusTotal
- PEStudio or Detect It Easy
- Strings / FLOSS
- Any.Run or Hybrid Analysis, if sandbox lookup is performed
- MITRE ATT&CK Framework
- Text editor for IOC documentation

Alert / initial evidence:

The initial evidence came from the 2025-05-12 Malware-Traffic-Analysis.net case describing a PureLogs infection from an email attachment. The case included an email artifact, PCAP infection traffic, an IOC text file, and malware/artifact files. Select email header evidence showed the message was received from

etsdc.com with source IP 185.222.57[.]74. The attachment invoice_10988.xz was identified as an archive that led to a disk image and extracted Windows executable.

Investigation steps:

1. Reviewed the Malware-Traffic-Analysis.net case page and identified the infection chain.
2. Collected the IOC, email, PCAP, and malware/artifact references provided by the case.
3. Reviewed the email evidence, including sender details, subject line, message ID, and attachment name.
4. Verified the attachment hash for invoice_10988.xz.
5. Identified the disk image invoice_10988.img extracted from the archive.
6. Identified the extracted executable KTMBE25040170.exe.
7. Checked file reputation using VirusTotal or similar reputation services.
8. Performed basic static analysis on the executable using PESTudio / Detect It Easy and strings utilities.
9. Opened the infection PCAP in Wireshark and filtered for suspicious destination IP, domain, and port activity.
10. Reviewed post-infection traffic to confirm external callback behaviour.
11. Extracted IOCs and mapped behaviour to MITRE ATT&CK techniques.
12. Documented impact, severity, and recommended remediation actions.

Findings:

Email and attachment evidence:

- Source domain: etsdc[.]com
- Source IP from received header: 185.222.57[.]74
- Email subject: RE: Urgent: Confirmation Required for Invoice & Down Payment Details
- Attachment: invoice_10988.xz

File / hash evidence:

- invoice_10988.xz SHA256: 341f58943626dec0cab58fbec4f7263125ec1ed75e0c97418cefe0ca23c6a25
- invoice_10988.img SHA256:
f757fc452dbb8eb564081d3decfdb31ec24fc4b91e22ee8088cb5884729cc99a
- KTMBE25040170.exe SHA256:
116c096a488f53b298d3bac99942770afd3d791ae376534f050e6e4642c2fbb4

File details:

- invoice_10988.xz was described as a ZIP archive using an .xz file extension.
- invoice_10988.img was described as an ISO 9660 CD-ROM filesystem image.
- KTMBE25040170.exe was described as a PE32 executable for Microsoft Windows.
- Post-infection file location: C:\Users\[username]\AppData\Roaming\Count.exe

VirusTotal score:

- A live VirusTotal lookup should be performed using the executable SHA256 hash. No specific VirusTotal score was published on the source case page, so the portfolio evidence should include a VirusTotal screenshot captured during analysis.

Basic static indicators:

- Windows executable extracted from an email-delivered archive/disk image chain.
- Suspicious file execution from a user profile AppData/Roaming location.
- Persistence mechanism referencing Count.exe through a Startup folder VBS script.
- Additional strings/imports should be documented after PEStudio or Strings analysis.

Sandbox / behavioural evidence:

- Malware family identified as PureLogs.
- Persistence observed through a VBS startup file launching Count.exe.
- Post-infection traffic was observed in the supplied PCAP.

Network callbacks:

- Destination IP: 176.65.144[.]169
- Domain: mxcnss.dns04[.]com
- Port: 7702
- Traffic type: encoded/encrypted TCP traffic

Indicators of Compromise:

- Source domain: etsdc[.]com
- Source IP: 185.222.57[.]74
- Email subject: RE: Urgent: Confirmation Required for Invoice & Down Payment Details
- Attachment: invoice_10988.xz
- Attachment SHA256: 341f58943626dec0cab58fbec4f7263125ec1ed75e0c97418cefe0ca23c6a25
- Disk image: invoice_10988.img
- Disk image SHA256: f757fc452dbb8eb564081d3decfdb31ec24fc4b91e22ee8088cb5884729cc99a
- Malware executable: KTMBE25040170.exe
- Executable SHA256: 116c096a488f53b298d3bac99942770afd3d791ae376534f050e6e4642c2fbb4
- Malware family: PureLogs
- Post-infection file: C:\Users\[username]\AppData\Roaming\Count.exe

- Persistence file: C:\Users\[username]\AppData\Roaming\Windows\Start Menu\Programs\Startup\Count.vbs
- C2 / callback domain: mxcnss.dns04[.]com
- C2 / callback IP: 176.65.144[.]169
- Port: 7702

MITRE ATT&CK mapping:

- T1566.001 - Phishing: Spearphishing Attachment
- T1204.002 - User Execution: Malicious File
- T1547.001 - Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder
- T1071 - Application Layer Protocol
- T1027 - Obfuscated Files or Information
- T1041 - Exfiltration Over C2 Channel (possible, depending on packet analysis)
- T1105 - Ingress Tool Transfer (possible, depending on observed payload transfer)

Impact assessment:

The case represents a successful email-based malware infection. The user received an invoice-themed attachment that led to an extracted executable associated with PureLogs. Post-infection traffic and persistence evidence indicate endpoint compromise risk. Because PureLogs is commonly associated with information stealing behaviour, potential impact includes credential exposure, browser/session data theft, persistence, continued outbound communication, and additional payload activity. Severity is assessed as High due to the confirmed post-infection traffic and persistence evidence.

Recommended actions:

- Isolate the affected endpoint from the network.
- Block mxcnss.dns04[.]com and 176.65.144[.]169 at DNS, proxy, firewall, and EDR controls.
- Search the environment for the listed file hashes and filenames.
- Remove persistence artifacts, including Count.vbs and Count.exe.
- Reset passwords for the affected user and any accounts accessed from the endpoint.
- Revoke active browser, email, VPN, and cloud sessions where possible.
- Run a full EDR/AV scan and collect forensic evidence before remediation if required.
- Review email-security controls for archive attachments and invoice-themed lures.
- Add SIEM detections for unusual execution from AppData/Roaming and outbound connections to suspicious infrastructure.
- Conduct phishing awareness training focused on invoice and payment-confirmation lures.

Conclusion:

This basic malware triage confirmed that the 2025-05-12 Malware-Traffic-Analysis.net PureLogs case is suitable for a GitHub-ready SOC portfolio project. The evidence supports a full investigation chain: email

attachment delivery, archive/disk-image extraction, Windows executable execution, persistence, post-infection traffic, IOC extraction, and MITRE ATT&CK mapping. The project demonstrates malware triage workflow, file hash analysis, static analysis planning, PCAP review in Wireshark, IOC documentation, and defensive remediation recommendations.